

Capturing traffic over Wi-Fi

Upload media immediately

Privacy Analysis of Augmented Reality Glasses

1 Introduction

This experiment captures network traffic over Wi-Fi during the interaction between an Android phone and the Ray Ban Meta glasses, more specifically while executing touch commands. Note that the photo/videos taken are manually imported onto the phone immediately after the capture. The aim of this data collection is to observe the network traffic generated during typical usage of the glasses for touch commands (to see how data flows between the device, its companion app and other apps). The goal would be to identify potential vulnerabilities or privacy breaches if data transmission is not securely protected. Furthermore, we can take a peek into whether any personally identifiable information or metadata can be taken during daily usage, therefore observe how user privacy may be affected.

2 Metadata

Time of experiment: Oct 16th 00:05

Duration of experiment: Roughly 13 minutes

IP addresses:

- Laptop - 192.168.0.215
- Redmi phone - 192.168.1.141
- Wi-Fi Network - 192.168.0.1
- ALFA Hotspot - 192.168.1.100/24
- Facebook servers - 157.240.x.x
- 35.186.x.x - Google cloud servers
- Meta CDN (content delivery network) - 157.240.17.17

3 Touch Commands used

- Keep idle for 1 minute (glasses off)
- Open Meta View
- Power On
- Pairing device
- Short click capture button to take a photo
- Upload image to phone
- Long click capture button to take a video recording
- Short click capture button to stop video recording
- Upload video to phone

- Long press capture button to take a full length video recording
- Upload video to phone
- Tap (and hold) on the side to play music
- Swipe forward to increase the volume
- Swipe backwards to decrease the volume
- Double tap to skip
- Triple tap to go previous
- Tap on the side to stop music
- Keep idle for 5 minutes (glasses on)

4 Summary

Protocol Usage

- TCP: For general communication with 7,336 packets.
- TLSv1.3: Secure connections with 5,208 packets.
- QUIC: For faster, more efficient connections with 1,207 packets.
- Other protocols like MDNS, DNS, and SSDP are also present but in smaller numbers.

IP communication

The phone (192.168.1.141) communicated frequently with mostly the following external IPs:

- 157.240.17.17 (Meta/Facebook): 4,876 packets
- 35.186.224.28 (Google Cloud): 583 packets
- 35.186.224.24 (Google Cloud): 297 packets
- 224.0.0.251 (Multicast): 500 packets, indicating internal broadcast traffic

Note

All packets: 15367

IPv4 - 96.4%

UDP - 13.5%

TCP - 82.6%

TLS - 34.8%

IPv4 -

Phone to 157.240.17.17 (owned by Meta CDN), 9486

Phone to 35.186.224.28 (owned by Google), 1155

Phone to 157.240.17.63 (owned by Instagram), 798

Phone to 35.186.224.24 (owned by Spotify, Google Cloud), 581

Phone to 224.0.0.251 (Multicast DNS address), 500

Phone to 157.240.17.3 (owned by Meta), 499

Phone to 142.250.203.106 (owned by Google Servers), 261

Phone to 1192.168.1.100 (ALFA Hotspot), 214

UDP -

Phone to 35.186.224.28 (owned by Google servers), 603

Phone to 35.186.224.24 (owned by Spotify, Google Cloud), 165

Phone to 157.240.17.17 (owned by Meta CDN), almost the rest

Phone to 157.240.17.3 (dedicated to Meta), very few

TCP -

Phone to 224.0.0.251 (Multicast DNS address), 500

Phone to 157.240.17.63 (owned by Instagram), near 800

Phone to 157.240.17.3 (owned by Meta), near 500

Phone to 1192.168.1.100 (ALFA Hotspot), the rest

Time Stamps

5 Analysis

- The traffic frequency graph (Figure 1) shows that the phone mostly interacts with the address 157.240.17.17 (Meta/Facebook) and Google Cloud IPs. This implies that the glasses and phone are heavily dependent on cloud services for functionality.
- There were multiple Meta-related IPs (157.240.17.17, 157.240.17.63) that were involved in the communication with the phone. This suggests that the glasses transmit a lot of data to Meta's servers, possibly for real-time processing or data syncing.

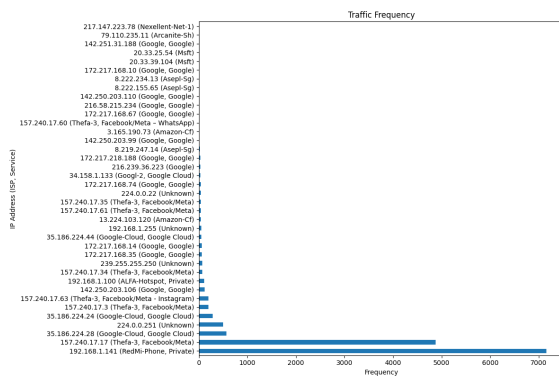


Figure 1: This graph shows the frequency of all traffic captured